What momentum says about impact

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect stopping impulse to average force without claiming a complete damage forecast.

SUCCESS CRITERIA - I can...

1. Compute momentum change during a stop.
2. Compare average force for different stopping times.
3. Name information beyond momentum needed for damage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “impact” and “momentum” to explain your model or result. Sentence starters are optional.

impact: a collision or strike between objects

momentum: mass times velocity, including direction

interval: the time or range between two stated values

01 Warm-up: bring back an earlier idea

Equal-mass carts have velocities +5 and -3 m/s before an ideal 1 D elastic collision.
Give both final velocities.

Word help: elastic = a collision that conserves total kinetic energy

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $p = mv$; $|F_{avg}| = |\Delta p|/\Delta t$.
Stopping force depends on stopping time as well as momentum. These quantities alone do not specify damage.

Worked reasoning: Two incoming objects have equal momentum 12 kg·m/s but masses 2 and 4 kg, so their speeds are 6 and 3 m/s. Momentum alone does not identify every possible consequence of stopping them.

Your turn: Draw equal momentum arrows and unequal velocity arrows.

Sentence starter: The ... in my model represents ... . Its label shows ... .



2 kg; 6 m/s 4 kg; 3 m/s

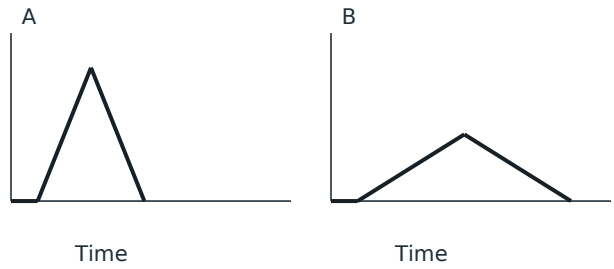
Marzano check - circle: ① ② ③

03 Read and interpret the model

Two equal-momentum objects stop in 0.1 s and 1 s. Which has greater average force magnitude, and by what factor?

Word help: magnitude = size of a quantity without its direction

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A 10 kg model target moves at 4 m/s and stops in 4 s. Complete its momentum row, then find the average stopping-force magnitude.
State the force direction.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: 1 2 3

05 Explain your reasoning

Why is a single stopping-force number not a prediction of buildings damaged by an asteroid?

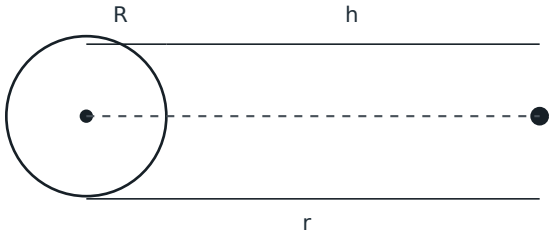
Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A planet has surface field 144 N/kg. At center distance 6 R, find field and the force on a 8 kg object.
Label the center distance, radius and field direction.

Sentence starter: My design shows ... . It meets the requirement by ... .



Marzano check - circle: 1 2 3

07 Exit check

If mass doubles while speed and stopping time stay fixed, how do momentum and average stopping force change?

Sentence starter: I notice ... . This supports ... because ... .

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2. Both objects stop in 0.5 s.
Compare their average stopping forces.
Explain why this does not establish identical peak forces or damage.

Sentence starter: Both ... . They differ in ... because ... .

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: 1 2 3